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Problem 4.12

For nitrogen molecules occupying a volume of 1 cm^3, calculate the translational energies corresponding to states for which (n_1, n_2, n_3) are (1,1,1), (100,100,100), and (1000,1000,1000).

Your response has been submitted. Engage and discuss with other learners below!

SD **Sandipan Dey** a few seconds ago

Translational Energy = E_{tr} = (h^2 / (8*m*V^(2/3))) * (n1^2 + n2^2+ n3^2)

m = 2*14*10^(-3) / (6.023 * 10^23)

E_{tr}(1,1,1) = 3.541514 x 10^{-38} J
E_{tr}(100,100,100) = 3.541514 x 10^{-34} J
E_{tr}(1000,1000,1000) = 3.541514 x 10^{-32} J

Like Reply Translate to English

GD **Gopal Dass** Learner 10 days ago

E 1,1,1=3.54×10 −22 J

E 100,100,100 =3.54×10 −18 J

E 1000,1000,1000 =3.54×10 −16 J

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AS **AMAN SIHAG** Learner 25 days ago

Good

Like Reply Translate to English

-- **César Peña** Learner a month ago

By using the expression $E = \frac{h^2}{8mL^2}\left(n_1^2 + n_2^2 + n_3^2\right)$ obtemos :

\[

E = \frac{h^2}{8mL^2} \left(n_1^2 + n_2^2 + n_3^2 \right)

\]

1. State (1, 1, 1):

- Energy: 3.54×10^{-38} J

- Energy: 2.21×10^{-19} eV

2. State (100, 100, 100):

- Energy: 3.54×10^{-34} J

- Energy: 2.21×10^{-15} eV

3. State (1000, 1000, 1000):

- Energy: 3.54×10^{-32} J

- Energy: 2.21×10^{-13} eV

Like Reply

SS **Shrinidhi Sharma** Learner a month ago

To calculate the translational energy corresponding to the quantum states , we use the expression for the translational energy levels of a particle in a three-dimensional box:

E_{n_1, n_2, n_3} = \frac{h^2}{8mL^2} \left(n_1^2 + n_2^2 + n_3^2\right)

Definitions:

: Planck's constant ()

: Mass of a nitrogen molecule ()

: Length of the box (side of the cubic volume)

Step 1: Relate to volume

The volume of the box is . For ,

L = V^{1/3} = (1 \times 10^{-6})^{1/3} = 1 \times 10^{-2}\, \mathrm{m}

Step 2: Energy calculation for each quantum state

The translational energy for a given state is:

E_{n_1, n_2, n_3} = \frac{h^2}{8mL^2} (n_1^2 + n_2^2 + n_3^2)

Substituting known constants:

\frac{h^2}{8mL^2} = \frac{(6.626 \times 10^{-34})^2}{8 \cdot (4.65 \times 10^{-26}) \cdot (1 \times 10^{-2})^2}

Calculation for each state:

1. For :

E_{1,1,1} = \frac{h^2}{8mL^2} \cdot (1^2 + 1^2 + 1^2)

2. For :

E_{100,100,100} = \frac{h^2}{8mL^2} \cdot (100^2 + 100^2 + 100^2)

3. For :

E_{1000,1000,1000} = \frac{h^2}{8mL^2} \cdot (1000^2 + 1000^2 + 1000^2)

Let's compute the numerical values.

Like Reply

JP **Jitendra Prajapat** Learner a month ago

This problem involves calculating the translational energies of nitrogen molecules (N2) in a cubic box of volume 1 cm^3, using the concept of quantum confinement.

Assumptions:

1. Nitrogen molecules are treated as point particles.

2. The box is cubic with sides of length L = 1 cm = 10^-2 m.

3. The molecules are non-interacting.

The translational energy of a particle in a 3D box is given by:

E(n1, n2, n3) = (h^2/8mL^2) * (n1^2 + n2^2 + n3^2)

where:

h = Planck's constant = 6.626 × 10^-34 J s

m = mass of a nitrogen molecule = 4.652 × 10^-26 kg (28.02 g/mol / Avogadro's number)

L = box length = 10^-2 m

(n1, n2, n3) = quantum numbers (positive integers)

Calculations:

(1, 1, 1)

E(1, 1, 1) = (h^2/8mL^2) * (1^2 + 1^2 + 1^2)

= (6.626^2 × 10^-68 / (8 × 4.652 × 10^-26 × 10^-4))

≈ 6.03 × 10^-22 J

≈ 3.76 × 10^-3 kT (at room temperature, kT ≈ 4.04 × 10^-21 J)

(100, 100, 100)

E(100, 100, 100) = (h^2/8mL^2) * (100^2 + 100^2 + 100^2)

= 6.03 × 10^-18 J

≈ 376 kT

(1000, 1000, 1000)

E(1000, 1000, 1000) = (h^2/8mL^2) * (1000^2 + 1000^2 + 1000^2)

= 6.03 × 10^-15 J

≈ 37600 kT

Results:

| (n1, n2, n3) | Energy (J) | Energy (kT) |

| --- | --- | --- |

| (1, 1, 1) | 6.03 × 10^-22 | 3.76 × 10^-3 |

| (100, 100, 100) | 6.03 × 10^-18 | 376 |

| (1000, 1000, 1000) | 6.03 × 10^-15 | 37600 |

Note: These energies are extremely small compared to thermal energies at room temperature, indicating that the translational motion of nitrogen molecules is largely classical.

Would you like me to explain any part of this calculation?

Like Reply

SM **Shashi Mohan** Learner a month ago

Amazing

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TC **Tanisha Chakrabarti** Learner a month ago

1

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SD **Sinjini Datta** Learner a month ago

1. For $(n_1, n_2, n_3) = (1, 1, 1)$: 6.08×10^{-23} J

2. For $(n_1, n_2, n_3) = (100, 100, 100)$: 6.08×10^{-19} J

3. For $(n_1, n_2, n_3) = (1000, 1000, 1000)$: 6.08×10^{-17} J

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TD **Tamal Dey** Learner a month ago

To calculate the translational energies of nitrogen molecules in different states, we use the **quantum mechanical expression** for the energy of a particle in a three-dimensional box, which is given by:

where:

- n1,n2,n3n_1, n_2, n_3n1,n2,n3 are the quantum numbers along the xxx, yyy, and zzz axes, respectively.
- ℏ\hbarℏ is the reduced Planck's constant, ℏ=1.0545718×10−34 J\cdotps\hbar = 1.0545718 \times 10^{-34} \, \text{J·s}ℏ=1.0545718×10−34J\cdotps.
- mmm is the mass of the nitrogen molecule, m=4.65×10−26 kgm = 4.65 \times 10^{-26} \, \text{kg}m=4.65×10−26kg.
- LLL is the characteristic length of the box, which is related to the volume of the box.
- kBk_BkB is the Boltzmann constant, kB=1.380649×10−23 J/Kk_B = 1.380649 \times 10^{-23} \, \text{J/K}kB=1.380649×10−23J/K.

Step 1: Calculate the characteristic length LLL

The volume of the box is given as V=1 cm3=1×10−6 m3V = 1 \, \text{cm}^3 = 1 \times 10^{-6} \, \text{m}^3V=1cm3=1×10−6m3. If the box is a cube, the side length of the cube LLL is:

So, the side length L=1 cm=1×10−2 mL = 1 \, \text{cm} = 1 \times 10^{-2} \, \text{m}L=1cm=1×10−2m.

Step 2: Calculate the energy for each state

The energy of the system in terms of quantum numbers n1,n2,n3n_1, n_2, n_3n1,n2,n3 is:

First, calculate the factor ℏ2π22mL2\frac{\hbar^2 \pi^2}{2mL^2}2mL2ℏ2π2. Substituting the known values:

This simplifies to:

Step 3: Calculate the energy for each state

For the state (n1,n2,n3)=(1,1,1)(n_1, n_2, n_3) = (1, 1, 1)(n1,n2,n3)=(1,1,1):

For the state (n1,n2,n3)=(100,100,100)(n_1, n_2, n_3) = (100, 100, 100)(n1,n2,n3)=(100,100,100):

For the state (n1,n2,n3)=(1000,1000,1000)(n_1, n_2, n_3) = (1000, 1000, 1000)(n1,n2,n3)=(1000,1000,1000):

En1,n2,n3=32kBT=ℏ2π22mL2(n12+n22+n32)E_{n_1, n_2, n_3} = \frac{3}{2} k_B T = \frac{\hbar^2 \pi^2}{2mL^2} \left(n_1^2 + n_2^2 + n_3^2\right)En1,n2,n3=23kBT=2mL2ℏ2π2(n12+n22+n32)L=(V)1/3=(1×10−6 m3)1/3=1×10−2 m=1 cmL = \left(V\right)^{1/3} = \left(1 \times 10^{-6} \, \text{m}^3\right)^{1/3} = 1 \times 10^{-2} \, \text{m} = 1 \, \text{cm}L=(V)1/3=(1×10−6m3)1/3=1×10−2m=1cmEn1,n2,n3=ℏ2π22mL2(n12+n22+n32)E_{n_1, n_2, n_3} = \frac{\hbar^2 \pi^2}{2mL^2} \left(n_1^2 + n_2^2 + n_3^2 \right)En1,n2,n3=2mL2ℏ2π2(n12+n22+n32)ℏ2π22mL2=(1.0545718×10−34)2π22(4.65×10−26)(1×10−2)2\frac{\hbar^2 \pi^2}{2mL^2} = \frac{(1.0545718 \times 10^{-34})^2 \pi^2}{2 (4.65 \times 10^{-26}) (1 \times 10^{-2})^2}2mL2ℏ2π2=2(4.65×10−26)(1×10−2)2(1.0545718×10−34)2π2ℏ2π22mL2≈1.416×10−22 J\frac{\hbar^2 \pi^2}{2mL^2} \approx 1.416 \times 10^{-22} \, \text{J}2mL2ℏ2π2≈1.416×10−22JE1,1,1=1.416×10−22×(12+12+12)=1.416×10−22×3=4.248×10−22 JE_{1, 1, 1} = 1.416 \times 10^{-22} \times (1^2 + 1^2 + 1^2) = 1.416 \times 10^{-22} \times 3 = 4.248 \times 10^{-22} \, \text{J}E1,1,1=1.416×10−22×(12+12+12)=1.416×10−22×3=4.248×10−22JE100,100,100=1.416×10−22×(1002+1002+1002)=1.416×10−22×3×104=4.248×10−18 JE_{100, 100, 100} = 1.416 \times 10^{-22} \times (100^2 + 100^2 + 100^2) = 1.416 \times 10^{-22} \times 3 \times 10^4 = 4.248 \times 10^{-18} \, \text{J}E100,100,100=1.416×10−22×(1002+1002+1002)=1.416×10−22×3×104=4.248×10−18JE1000,1000,1000=1.416×10−22×(10002+10002+10002)=1.416×10−22×3×106=4.248×10−16 JE_{1000, 1000, 1000} = 1.416 \times 10^{-22} \times (1000^2 + 1000^2 + 1000^2) = 1.416 \times 10^{-22} \times 3 \times 10^6 = 4.248 \times 10^{-16} \, \text{J}E1000,1000,1000=1.416×10−22×(10002+10002+10002)=1.416×10−22×3×106=4.248×10−16J

Like Reply

SH **SYED MD AMAANUL HAQUE** Learner a month ago

The translational energy of a particle in a three-dimensional box of dimensions Lx, Ly, and Lz is given by:

E = (h^2 / 8m) * (n1^2/Lx^2 + n2^2/Ly^2 + n3^2/Lz^2)

Where:

* h is Planck's constant (6.626 x 10^-34 J s)

* m is the mass of the particle (in this case, the mass of a nitrogen molecule, which is approximately 4.65 x 10^-26 kg)

* n1, n2, and n3 are quantum numbers corresponding to the particle's motion in the x, y, and z directions, respectively.

* Lx, Ly, and Lz are the dimensions of the box.

Calculations

For a 1 cm^3 box, we can assume it's a cube with sides of length L = 1 cm = 0.01 m.

State (1,1,1):

E = (6.626 x 10^-34 J s)^2 / (8 * 4.65 x 10^-26 kg) * (1^2 + 1^2 + 1^2) / (0.01 m)^2

= 1.06 x 10^-21 J

State (100,100,100):

E = (6.626 x 10^-34 J s)^2 / (8 * 4.65 x 10^-26 kg) * (100^2 + 100^2 + 100^2) / (0.01 m)^2

= 1.06 x 10^-17 J

State (1000,1000,1000):

E = (6.626 x 10^-34 J s)^2 / (8 * 4.65 x 10^-26 kg) * (1000^2 + 1000^2 + 1000^2) / (0.01 m)^2

= 1.06 x 10^-13 J

Summary: